

Assignment: NET Centric Computing

Course Code: CSC378

Topic: Unit 5 - Working with Database (Theory & Practical)

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Due Date: 24 August

Submission Instructions:

Please complete the following questions by combining theoretical explanations with practical implementations. For practical components, include required commands, code snippets, database design details, and screenshots of application execution and database results.

Questions:

1. Database Operations Using ADO.NET Classes

Explain the basic concepts of ADO.NET and describe the roles of Connection, Command, Reader, and Adapter classes. Explain how these classes work together to perform database operations.

Develop an ADO.NET application to connect with a database and perform data retrieval and manipulation operations.

Perform the following tasks:

1. Create a database and design a table (such as Student, Faculty, Book, or Movie) with appropriate fields.
2. Establish a database connection using the ADO.NET Connection class.
3. Implement SQL operations using the Command class:
 - Insert records.
 - Update records.
 - Delete records.
 - Retrieve records.
4. Use the Reader class to read and display records from the database.
5. Use the Adapter class and DataSet/DataTable to retrieve and manage disconnected data.
6. Implement sample operations such as:
 - Insert Faculty details.
 - Retrieve records based on conditions.

- Update specific records.
- Delete records based on criteria.

7. Document:

- Database connection process.
- Usage of Connection, Command, Reader, and Adapter classes.
- SQL queries used.
- Output screenshots.

2. Database Application Development Using Entity Framework Core

Explain the concept of Entity Framework Core (EF Core) and Object-Relational Mapper (ORM). Discuss the advantages of EF Core over traditional ADO.NET. Explain database providers, data models, and data context in EF Core.

Develop an ASP.NET Core application using Entity Framework Core to perform database operations.

Perform the following tasks:

1. Add Entity Framework Core to an ASP.NET Core application.
2. Select and configure an appropriate database provider:
 - SQL Server / SQLite / MySQL.
3. Create database models (entities), such as:
 - Employee.
 - Product.
 - Student.
4. Create and configure the Data Context (DbContext) class.
5. Perform CRUD operations using Entity Framework Core:
 - Create: Insert new records into the database.
 - Read: Retrieve records based on conditions.
 - Update: Modify existing records.
 - Delete: Remove records from the database.
6. Implement queries using EF Core to demonstrate:

- Filtering records.
- Updating selected records.
- Deleting records based on conditions.

7. Demonstrate database communication through ORM without writing complex SQL queries.

8. Document:

- EF Core architecture.
- ORM working mechanism.
- Data model and Data Context design.
- CRUD implementation.
- Database output screenshots.